

3.2.)

a) $r_1=1$ $r_2=x$ $r_3=x^2$, chei ON bazi

$$\bullet w_1 = \frac{r_1}{\|r_1\|} \quad \|r_1\|^2 = \int_1^5 1 \cdot 1 = 4 \\ \Rightarrow \|r_1\| = 2$$

$$w_1 = \frac{1}{2}$$

$$\bullet w_2: \quad w_2 = \langle w_1, r_2 \rangle \cdot w_1 = \int_1^5 \frac{1}{2} x \cdot \frac{1}{2} = 6 \cdot \frac{1}{2} = 3 \\ r_2 - w_2 = x - 3 \quad \|x-3\|^2 = \int_1^5 (x-3)^2 = \frac{16}{3} \\ \Rightarrow \|x-3\| = \frac{4}{\sqrt{3}}$$

$$w_2 = \frac{x-3}{\frac{4}{\sqrt{3}}} = \frac{\sqrt{3}}{4} (x-3)$$

$$\bullet w_3: \quad w_{31} = \langle w_1, r_3 \rangle w_1 = \int_1^5 \frac{1}{2} x^2 \cdot \frac{1}{2} = \frac{62}{3} \cdot \frac{1}{2} = \frac{31}{3} \\ w_{32} = \langle w_2, r_3 \rangle w_2 = \int_1^5 \frac{\sqrt{3}}{4} (x-3) \cdot x^2 \cdot \frac{\sqrt{3}}{4} (x-3) \\ = 8\sqrt{3} \cdot \frac{\sqrt{3}}{4} (x-3) = 6x - 18$$

$$r_3 - w_{31} - w_{32} = x^2 - \frac{31}{3} - 6x + 18 \\ = x^2 - 6x + \frac{23}{3}$$

$$\|r_3 - w_{31} - w_{32}\|^2 = \int_1^5 \left(x^2 - 6x + \frac{23}{3}\right)^2 = \frac{256}{45} \\ \Rightarrow \|r_3 - w_{31} - w_{32}\| = \frac{16}{3\sqrt{5}}$$

$$w_3 = \frac{x^2 - 6x + \frac{23}{3}}{\frac{16}{3\sqrt{5}}} = \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3}\right)$$

$$\Rightarrow (w_1, w_2, w_3) = \left(\frac{1}{2}, \frac{\sqrt{3}}{4}(x-3), \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3}\right)\right)$$

b) projekcija $\sin x$ na $\text{Span}\{1, x, x^2\}$
oznacimo je $\vec{w}$

maim z a) ON bazi w_1, w_2, w_3

$$\Rightarrow \vec{w} = \langle w_1, \sin x \rangle w_1 + \langle w_2, \sin x \rangle w_2 + \langle w_3, \sin x \rangle w_3$$

$$\langle w_1, \sin x \rangle = \int_1^5 \frac{1}{2} \sin x \approx 0,128$$

$$\langle w_2, \sin x \rangle = \int_1^5 \frac{\sqrt{3}}{4} (x-3) \cdot \sin x \approx -1,4932$$

$$\langle w_3, \sin x \rangle = \int_1^5 w_3 \cdot \sin x \approx -0,1252$$

$$\vec{w} = \frac{1}{2} 0,128 - 1,493 \left(\frac{\sqrt{3}}{4} (x-3)\right) - 0,125 \left(\frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3})\right)$$

$$= 0,064 - 0,646x + 1,439 - 0,052x^2 + 0,314x - 0,402$$

$$= -0,052x^2 - 0,332x + 1,601$$

$$c) \quad \|\sin(x) - (ax^2 + bx + c)\| = ?$$

$$\|\sin(x) - (ax^2 + bx + c)\|^2 = \langle \sin x - (w), \sin x - (w) \rangle$$

$$= \int_1^5 \left(\sin x + 0,052x^2 + 0,332x - 1,601\right)^2 dx \approx 0,0138$$

$$\|\sin x - \vec{w}\| = \sqrt{0,0138} = 0,117$$